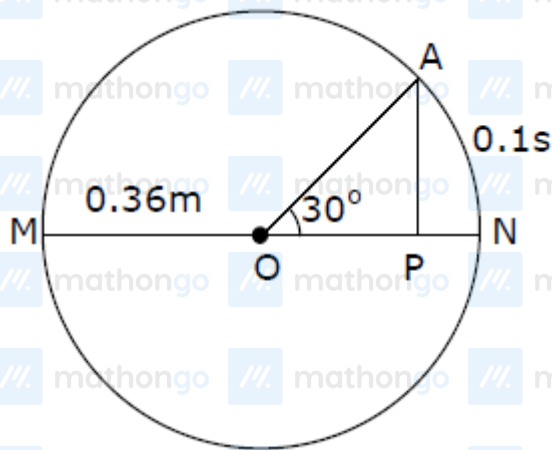


Q1: 25 Feb (Shift 1) - Numerical

A small bob tied at one end of a thin string of length 1 m is describing a vertical circle so that the maximum and minimum tension in the string are in the ratio 5: 1 . The velocity of the bob at the highest position is m/s. (take $g = 10 \text{ m/s}^2$)

Q2: 25 Feb (Shift 2) - Single Correct

The point A moves with a uniform speed along the circumference of a circle of radius 0.36 m and cover 30° in 0.1s. The perpendicular projection 'P' from 'A' on the diameter MN represents the simple harmonic motion of 'P'. The restoration force per unit mass when P touches M will be:



(1) 100 N

(2) 50 N

(3) 9.87 N

(4) 0.49 N

Q3: 26 Feb (Shift 1) - Single Correct

A particle is moving with uniform speed along the circumference of a circle of radius R under the action of a central fictitious force F which is inversely proportional to R^3 . Its time period of revolution will be given by :

(1) $T \propto R^{\frac{5}{2}}$ (2) $T \propto R^2$ (3) $T \propto R^{\frac{4}{3}}$

Questions with Answer Keys

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 (4) $T \propto R^{\frac{3}{2}}$					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					

Answer Key

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Q1 (5)

Q2 (3)

Q3 (2)

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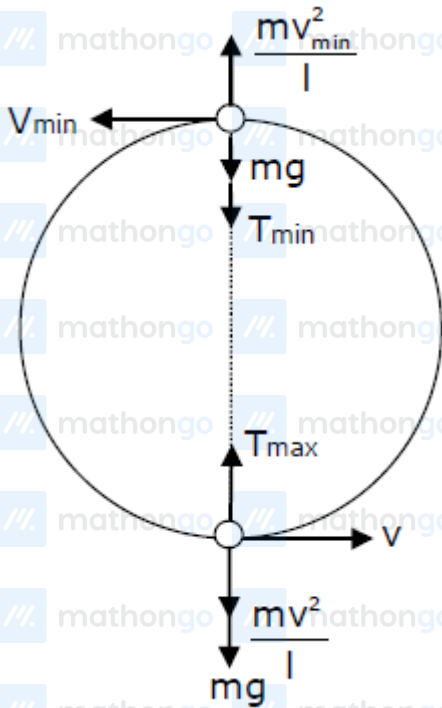
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Q1 (5)



by conservation of energy, $v_{\min}^2 = V^2 - 4gl$ (1)

$$T_{\max} = mg + \frac{mv^2}{l} \dots (2)$$

$$T_{\min} = \frac{mv_{\min}^2}{l} - mg \dots (3)$$

from equation (1) and (3)

$$T_{\min} = \frac{m}{l} (v^2 - 4gl) - mg$$

$$\frac{T_{\max}}{T_{\min}} = \frac{\frac{v^2}{l} + g}{\frac{v^2}{l} - 5g}$$

$$\frac{5}{1} = \frac{\frac{v^2}{l} + 10}{\frac{v^2}{l} - 50}$$

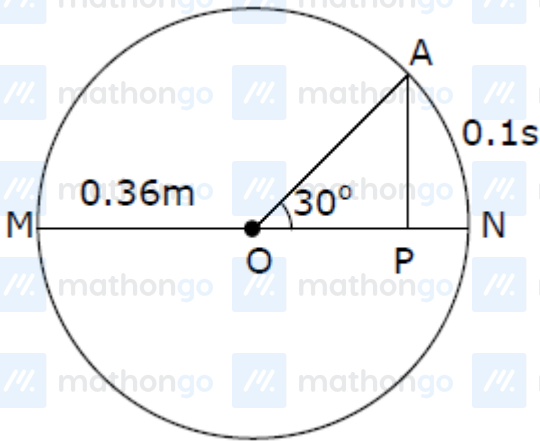
$$5v^2 - 250 = v^2 + 10$$

$$v^2 = 65 \dots (4)$$

from equation (4) and (1) $v_{\min}^2 = 65 - 40 = 25$

$$v_{\min} = 5$$

Q2 (3)



The point a covers 30° in 0.1 sec.

Means $\frac{\pi}{6} \rightarrow 0.1\text{sec.}$

$$1 \rightarrow \frac{0.1}{\frac{\pi}{6}}$$

$$2\pi \rightarrow \frac{0.1 \times 6}{\pi} \times 2\pi$$

$$T = 1.2\text{sec}$$

$$\text{We know that } \omega = \frac{2\pi}{T}$$

$$\omega = \frac{2\pi}{1.2}$$

$$\text{Restoration force } (F) = m \omega^2 A$$

$$\text{Then Restoration force per unit mass } \left(\frac{F}{m}\right) = \omega^2 A$$

$$\left(\frac{F}{m}\right) = \left(\frac{2\pi}{1.2}\right)^2 \times 0.36$$

$$\cong 9.87 \text{ N}$$

Q3 (2)

$$F \propto \frac{1}{R^3}$$

$$F = \frac{K}{R^3}$$

$$\frac{mv^2}{R} = \frac{K}{R^3}$$

$$m(\omega R)^2 = \frac{K}{R^2}$$

$$m\omega^2 R^2 = \frac{K}{R^2}$$

$$\omega^2 = \frac{K}{m} \left(\frac{1}{R^4}\right)$$

Hints and Solutions

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$$\left(\frac{2\pi}{T}\right)^2 \propto \frac{1}{R^4}$$

$$\frac{4\pi^2}{T^2} \propto \frac{1}{R^4}$$

$$T \propto R^2$$